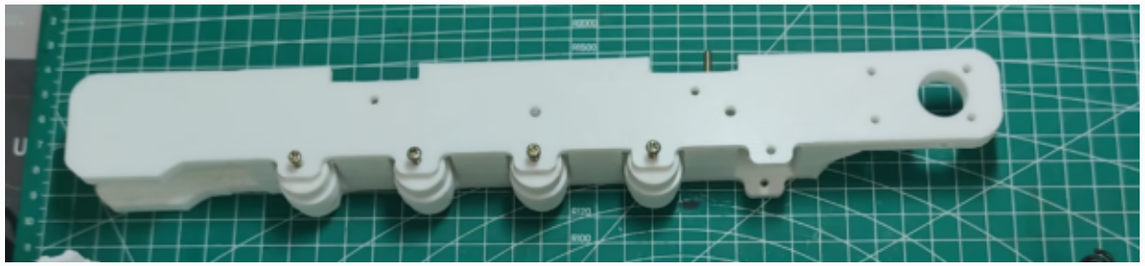
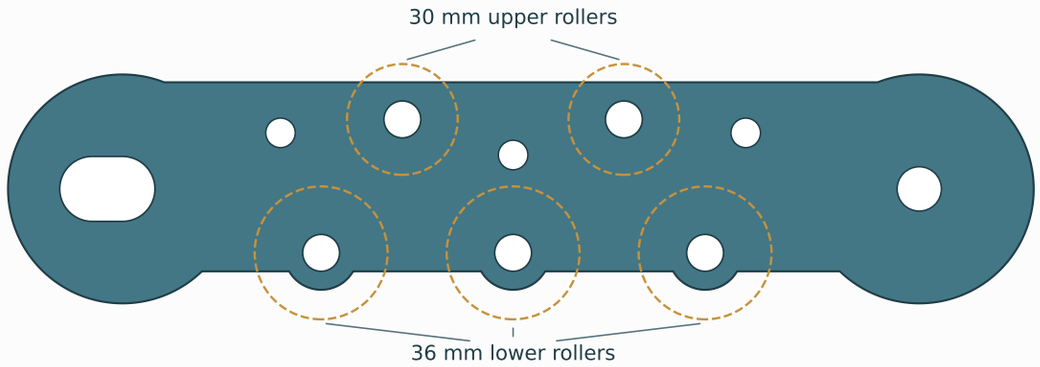
J8 / Continuous side frames on both sides

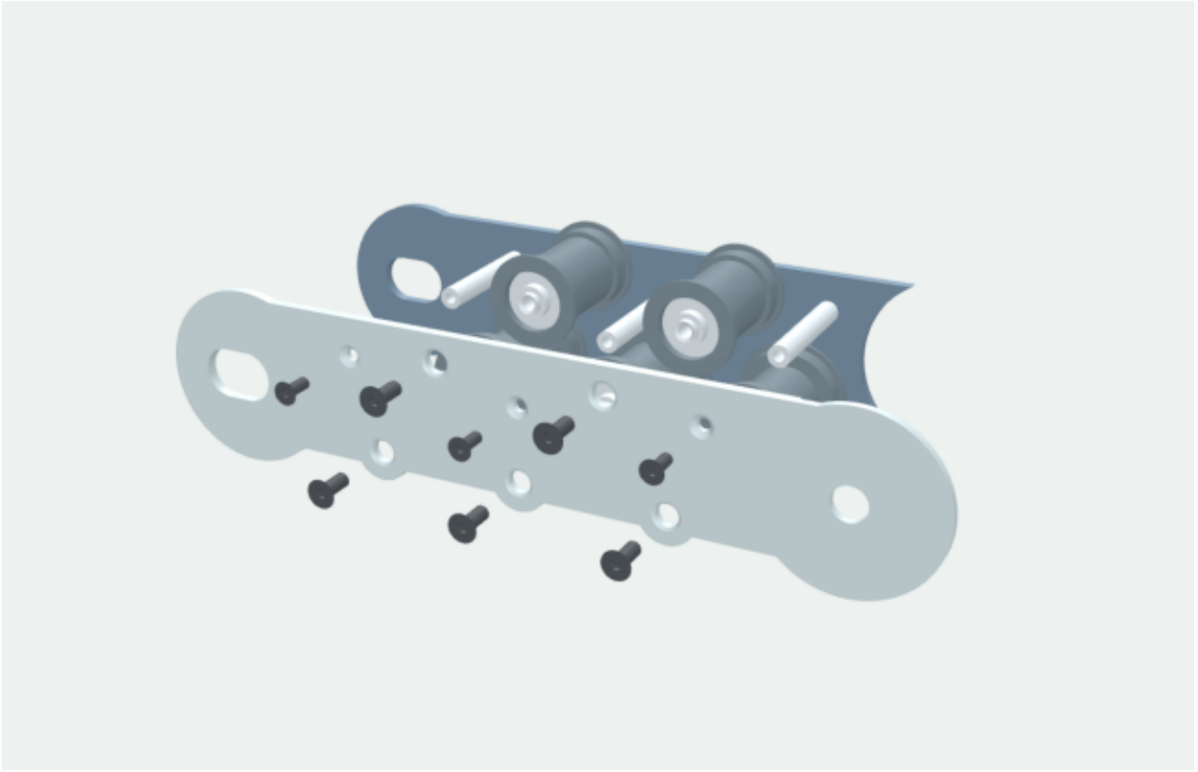
Your reference layout, adapted to the retained motor, track dimensions and five roller positions.



Your reference: a broad continuous web with wheel mounting points.



J816 outer side plate / actual CAD profile. Dashed lines show roller diameters.



One continuous outer side

The full-length web joins the wheel mounts.
Five axle holes match the inner side.
Three cross ties connect the two plates.
The complete outer plate removes for service.

Keep the end mechanisms free

Drive access: 12 mm opening.
Idler opening follows the original travel.
Drive bearings and sliding yoke retained.
Motor-side plate has the necessary drive clearance.

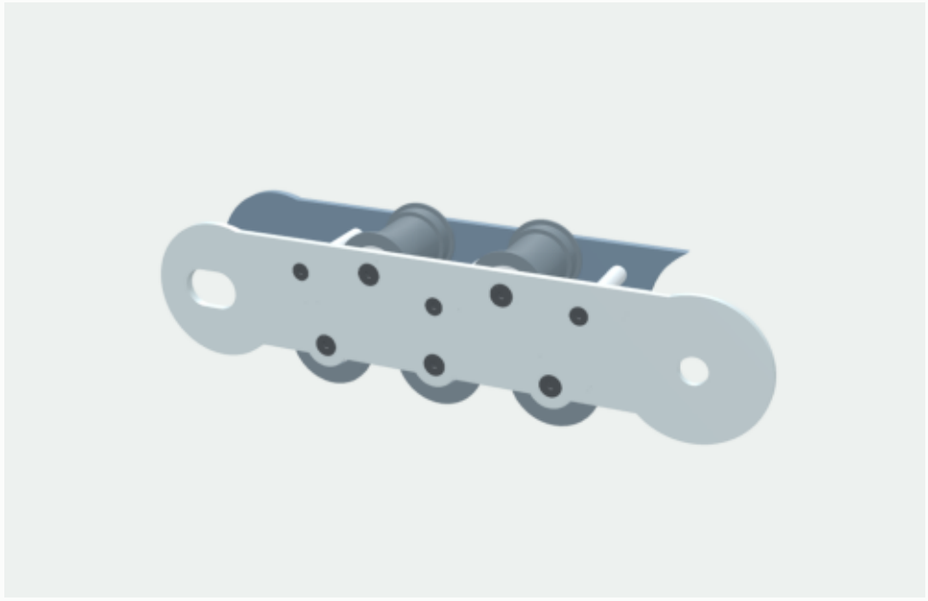
Same overall package

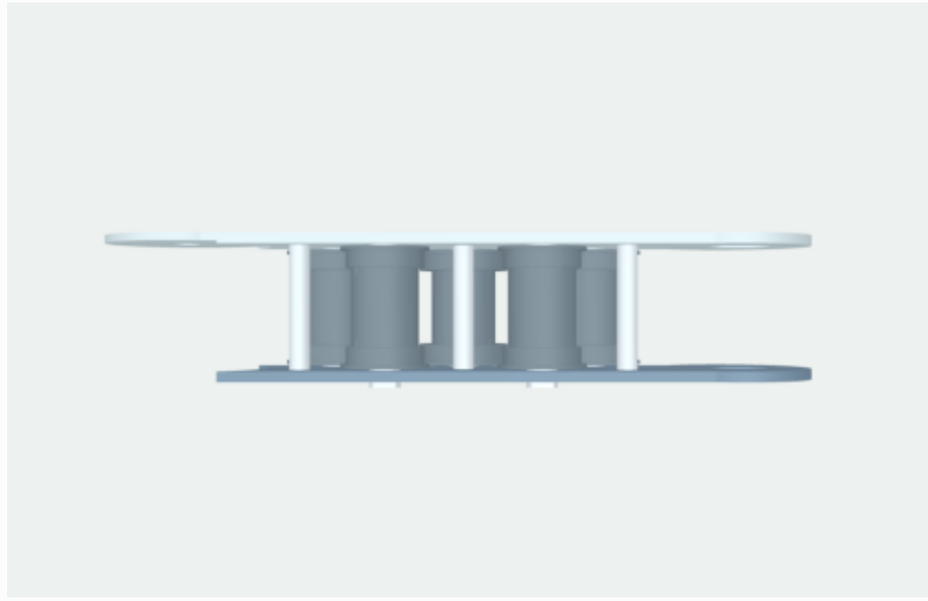
308.44 x 101 x 91.14 mm.
Original motor and chassis mounting points.
One-piece protective housing retained.

Both sides support the small-wheel axles. Current holes fit the retained 30 / 36 mm rollers; new sizes need a fresh track-clearance check.

Continuous plates with matched axle positions

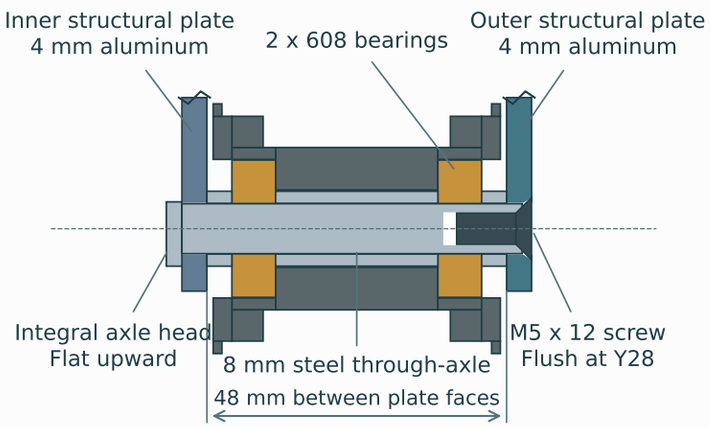
The removable cheek closes the support structure. It must be fitted for operation.





Inner cheek / J815

4 mm seat: Y=-28 to -24.
Axle head bears against the metal cheek.
Frame access pocket clears the rear head.



Outer cheek / J816

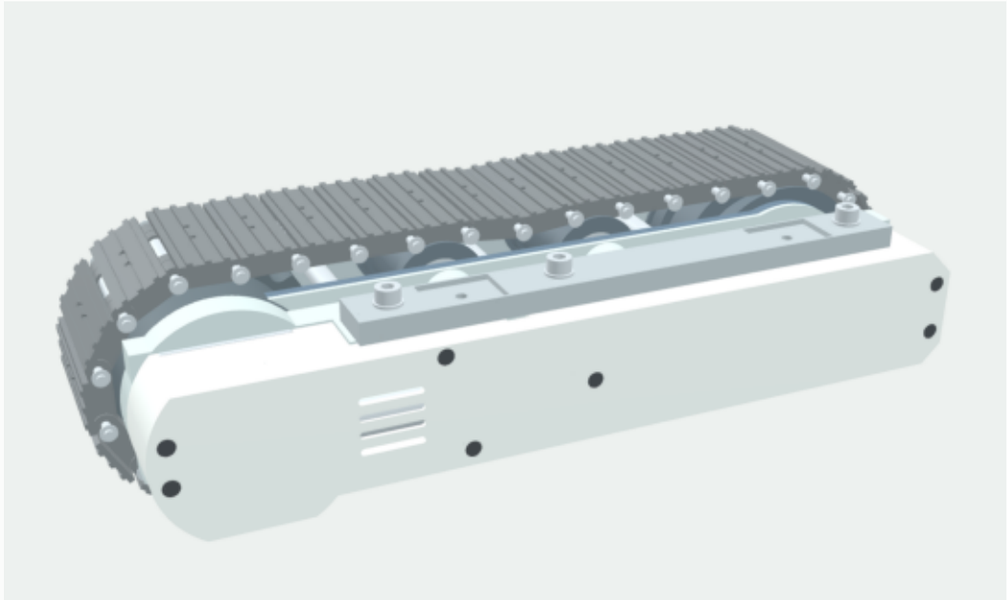
2.5 mm straight shaft engagement: Y24 to 26.5.
The remaining plate depth contains the countersink.
Screw head finishes flush at Y28.

Check fits before loaded use

Machine the plates as a matched pair.
Match five bearing stacks to the 48 mm cross ties.
Verify free rotation and 1 mm flange clearance per side.
Finish journals, blind threads and fastener seats.
No payload rating is established.

Service access and operation

New J8 CAD, with the unchanged J6 unloaded motion trace replayed in the video.



Nominal geometry checks

151 changed-part collision pairs clear.
Original overall bounds retained.
12 outer-plate withdrawal positions checked.
45 inboard axle-removal samples clear.
30 wheel-removal samples clear.
15 sampled motion poses clear.

Service sequence

1. Remove housing for rear fastener access.
2. Remove eight outer screws, then the outer plate.
3. Open/remove chain before withdrawing wheels.
4. Withdraw a bare axle inboard, head flat upward.
5. Refit both plates and check free rotation.

Video provenance

J8 geometry; reused J6 motion, not a new simulation.
Generic servo target: +/-20 rpm.
Nominal chain speed: 80 mm/s.
Unloaded fixed frame; not physical footage.
No motor electrical or vehicle load model.

Both continuous plates carry the small-wheel axle loads. The one-piece PA12 housing remains separate, with 2.5 mm nominal walls and a local 1 mm motor skin.

Actual X-frame / left motor fit, screw preload, plate bearing stress, printed-frame creep and loaded endurance remain unverified.